



MittiAI

The agronomist in every farmer's pocket

Google Cloud "Build with AI: Code for Communities" · Track 4 — AgriN & Regenerative Agricultural Intelligence

Gemini multimodal AI

Real Indian open data

6 languages · voice-first

THE PROBLEM

100M+ smallholders fight crop disease **with guesswork**

30–40%

crop yield lost to pests & disease
annual losses worth ₹1.4+ lakh crore

1 : 1,000+

agronomists vs. farmers
extension officers are unreachable at scale

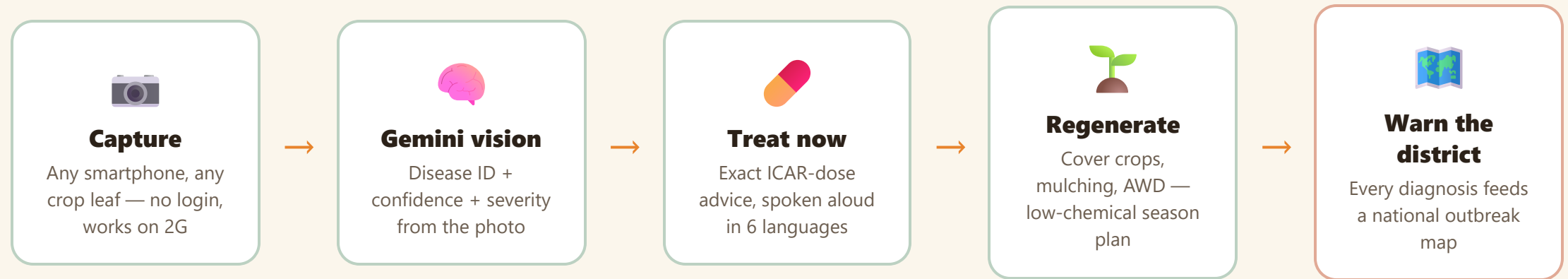
Days

typical diagnosis lag
by the time disease is identified, it has spread

Disease pressure is also a **national problem**: outbreaks cross district borders silently — no early-warning layer exists between the field and the agriculture department.

THE SOLUTION

Photo in. Diagnosis, treatment & a regenerative plan out.



An **interoperable digital agriculture network** — exactly the Track-4 brief: localized advisories + disease diagnostics, built as a federated platform.

Four screens, one network

/diagnose

Leaf photo → Late Blight · 93% confidence · HIGH severity ·
Mancozeb 2.5 g/L + regenerative alternatives · spoken readout

/advisory

Soil Health Card + 12-week NDVI sparkline + 14-day rain view →
Gemini sowing window, cover crop & 10-week calendar

/chat — Sahayak

Voice-in, voice-out chat in हिंदी, मराठी, தமிழ், తెలుగు, বাংলা,
English — grounded on the farmer's plot data

/dashboard

12-district outbreak map, early-warning queue, Gemini-drafted
officer bulletins, Agmarknet price ticker

Gemini does the heavy lifting **four ways**

Multimodal vision diagnosis

Gemini vision reads field photos (blurry, low-light tolerant) → structured JSON via **responseSchema**: disease, confidence, severity, treatment.

Data-fused advisory engine

Soil nutrients + NDVI trend + 14-day forecast are composed into one prompt → structured regenerative calendar JSON.

Multilingual voice loop

Web Speech STT → Gemini answers in the farmer's language → TTS readout. No literacy barrier.

Officer bulletin drafting

Gemini composes district early-warning bulletins from live outbreak facts — SMS/IVR ready.

Every call: timeout + **demo-mode fallback** — the product never hard-fails on stage.

Grounded in India's public data layer



Soil Health Card

data.gov.in — N, P, K, OC, pH profiles
drive soil-fix advice



ISRO Bhuvan / Sentinel-2

12-week NDVI canopy trends per plot —
stress detection



IMD district forecasts

14-day rain/temp/humidity → irrigation
& spray windows



Agmarknet

Mandi modal prices with 7-day change
ticker



ICAR / PlantVillage KBs

Disease profiles, symptoms, ICAR-
recommended doses



Surveillance patterns

ICAR pest-surveillance-style district
reports seed the live feed

Modular feed seam: components never touch data directly — real APIs (IMD, Bhuvan, Earth Engine) plug in without UI changes.

THE FLYWHEEL

Every diagnosis makes **the whole network smarter**

1. Farmer photographs

Diagnosis logged with district, crop, disease, timestamp.

2. Map aggregates

District clusters surface rising outbreaks before they epidemic.

3. Officer acts

Gemini-drafted bulletins dispatch targeted advisories — farmers get SMS/IVR in their language.

This is the shift from **individual advice** to **population-level plant health intelligence** — the Digital Public Good angle.

BENEFICIARIES

Who wins, and how



Smallholder farmers

30-sec diagnosis in their language → less crop loss, lower chemical spend, higher income. Voice-first = no literacy barrier.



Agriculture departments

Real-time district outbreak visibility → targeted, cheaper interventions instead of blanket responses.



Food system

Early warning protects national yield — directly serves food-security mandates.

A district row is all it takes

- **Add a district = add a state.** Feed schemas are portable objects (lat/lon, crop, disease, severity).
- **BRICS swap-ins map 1:1** — Brazil INPE satellite feeds, South Africa weather data, Russia FCDC bulletin schemas plug into the same seam.
- **2G-tolerant, offline-friendly** — seeded feeds keep rural deployments alive when networks drop.
- **DPG path** — modular open-source architecture fits Digital Public Good certification.

Public infrastructure, self-sustaining economics



State DPI contracts

License the advisory + early-warning layer to state agriculture departments.



FPO partnerships

Farmer-producer orgs bundle MittiAI advisory as a member service.



Insurance data

Anonymized outbreak intelligence prices crop-insurance risk more fairly.

From prototype to **state pilot**

Now

Gemini diagnosis, advisory, 6-language voice chat, outbreak dashboard, officer dispatch — live and deployed.

+30 days

Live IMD/Bhuvan API wiring, WhatsApp bot entry point, missed-call IVR diagnosis for feature phones.

+90 days

One-state pilot with an agriculture department; 10,000-farmer cohort; publish outbreak bulletin #1.



From guesswork to precision — **in every farmer's language.**

Live demo · github.com/arz1man/mittiai · Built with the Gemini API

Team MittiAI · arz1man — Google Cloud "Build with AI: Code for Communities"